

CURRICULUM VITAE (CV)

Jin-Bin Im – Postdoctoral Researcher

Chalmers University of Technology, Division of Construction Management and Engineering

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SUMMARY

- **Expertise** – AI-driven human-centered architectural design, affective spatial computing, occupant experience assessment, extended reality, physiological responses, BIM-integrated design automation
- **Interdisciplinary background** – Grounded in Interior Architecture Design, advancing engineering research that bridges spatial design thinking with computational methods and biometric-driven design.
- **Publication** – I have published multiple peer-reviewed articles in high-impact international journals in the fields of architectural design, computational design, and the built environment, including *Architectural Science Review* (A&HCI, top 2.58%), *Building and Environment* (IF 7.6), *Architectural Engineering and Design Management* (IF 2.7), and *Journal of Management in Engineering* (IF 7.0).
- **Academic Service** – I have served the academic communities as a reviewer of high impact journals, such as, *Building and Environment*, *Architectural Engineering and Design Management*, and *Journal of Management in Engineering*.

RESEARCH INTERESTS

- **AI-Driven Architectural Design**
 - Affective computing for spatial design generation
 - Client requirements-driven automated design optimization
 - Evidence-based decision support systems for personalized architectural design
- **Immersive Virtual Environment for End-User Engagement in Design Review**
 - XR (VR/MR) platforms for immersive co-design and real-time spatial experience with clients
 - Human-space interaction and occupant behavior analysis in dynamic architectural environments
 - Sensor-driven evidence-based design for health and well-being assessment
- **Human-Centered Design Computing**
 - End-to-end pipelines from multimodal data acquisition to BIM-linked structured output
 - Natural language processing for translating user requirements into design parameters
 - Interoperable digital design frameworks supporting downstream lifecycle analysis

EDUCATION

Hanyang University

Mar. 2020–Aug. 2026

M.S/Ph.D. in Architectural Engineering, Construction Management

Dissertation title: “Distribution-aware Affective Requirement Realignment for User Interactive Spatial Design Revision”

Hanyang University

Mar. 2014–Feb. 2020

B.S. in Interior Architecture Design

EXPERIENCE

Postdoctoral Researcher , Chalmers University of Technology	Sep. 2026–present
Research Assistant , Hanyang University, <i>Dept. of Architectural Engineering</i>	Mar. 2020–Aug. 2026
• <i>Development of Multi-functional Construction Robots to Enhance Productivity in Apartment Housing Construction</i>	Apr. 2026–Aug. 2026
• <i>Development of Artificial Intelligence-based Real-Time Construction Quality Management System.</i>	May. 2024–Aug. 2026
• <i>Study on Quality Control of Educational Facilities using Virtual Environments.</i>	Oct. 2023–Oct. 2024
• <i>Development of Decision-making Framework to Support Personalized Spatial Composition based on Temperament, Perception and Bio-markers Applying Digital Twin.</i>	Sep. 2021–Feb. 2024
• <i>Construction Quality Control using Digital Twin Technology.</i>	Apr. 2021–Dec. 2022
• <i>Establishment of Standard Guidelines for Daycare Centers in Police Stations.</i>	Jun–Sep. 2021
Internship , SIFC Property Korea YH	Nov. 2019–May. 2020
• <i>Property management for IFC Seoul, particularly an IFC Mall (85,444m² GFA).</i>	
Research Intern , Hanyang University, <i>Dept. of Interior Architecture Design</i>	May–Jul. 2019
• <i>Spatial Preference Modeling based on Immersive Virtual Environment and Choice Modeling Theory.</i>	

TEACHING

Instructor , Korea University of Technology and Education	2026
• <i>AI-based Optimization Technology for Architectural Design</i>	
Teaching Fellow , Hanyang University, <i>Dept. of Architectural Engineering</i>	2024–2025
• <i>PBL4004 Architectural Engineering Capstone PBL</i>	Fall 2025
• <i>ENE4044 Computer Engineering</i>	Spring 2024, 2025

SERVICE

Reviewer , 23 rd Smart and Sustainable Built Environment (SASBE)	2026
Reviewer , <i>Virtual Reality</i>	2026–present
Reviewer , <i>Acta Psychologica</i>	2026–present
Reviewer , <i>Results in Engineering</i>	2026–present
Reviewer , <i>Journal of Management in Engineering</i>	2026–present
Reviewer , <i>Architectural Engineering and Design Management</i>	2025–present
Reviewer , <i>Building and Environment</i>	2025–present

PATENT

- *South Korea Patent, No. 10-2421360, “System for Generating 3D LiDAR Scanning Position.”* Jul. 2022

• REFERRED JOURNAL

- [J15] Zhang, E., **Im, J. B.**, Lee, K. T., & Kim, J. H.* (2026). Visual quality features in immersive virtual environments: A systematic review of their impact on user experience. *Virtual Reality*, Springer Nature. [SCIE, Q1, 11.2%, IF 5.4] (accepted)
<https://doi.org/10.1007/s10055-026-01433-z>
- [J14] Hong, R. L., Yoon, J. H., **Im, J. B.**, & Kim, J. H.* (2026). Prediction of Actual Safety and Health Management Expenses for Water Supply System Construction Projects in Korea through Machine Learning. *Journal of Civil Engineering and Management*, Vilnius Tech Press. [SCIE, Q1, 14.05%, IF 3.7] (accepted)
- [J13] Hong, R. L., **Im, J. B.**, Kang, H. G., & Kim, J. H.* (2026). Prediction of structural frame works safety and health management expenses in apartment construction projects by introducing interpretable machine learning. *Engineering, Construction and Architectural Management*, Emerald. [SCIE, Q1, 22.3%, IF 3.6]
<http://doi.org/10.1108/ECAM-07-2025-1073>
- [J12] Joo, M., **Im, J. B.**, Lee, K. T., & Kim, J. H.* (2026). Comparative Analysis of Work Performance and Physiological Response Based on Office Types and Indoor Space Design. *Architectural Engineering and Design Management*, Taylor & Francis, 22(2), 365–381. [SCIE, Q2, 34.3%, IF 2.7]
<https://doi.org/10.1080/17452007.2025.2490958>
- [J11] Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2026). Effects of Human Perception Dimensions Constituting Individual Temperament and Experience on Indoor Space Composition Decision-making Presented in a Mixed-reality Environment. *Architectural Science Review*, Taylor & Francis, 69(3), 314–329. [A&HCI, Q1, 2.58%, IF 1.8]
<https://doi.org/10.1080/00038628.2025.2510300>
- [J10] Lee, K. T., Shin, S. H., **Im, J. B.**, Choi, H., & Kim, J. H.* (2026). Integrating AI-Enhanced and Conventional Market Selection Approaches for the Construction Industry: Insights from Indian Firms. *Journal of Civil Engineering and Management*, Vilnius Tech Press, 32(3), 456–474. [SCIE, Q1, 14.05%, IF 3.7]
<https://doi.org/10.3846/jcem.2026.26811>
- [J9] Lee, K. T., **Im, J. B.**, Hong, R. L., & Kim, J. H.* (2026). Enhancing Retrieval-Augmented Generation Performance through Network Analysis of Question Types in Public Procurement Law. *Journal of Management in Engineering*, ASCE, 42(4), 04026014. [SCIE, Q1, 4.35%, IF 7.0]
<https://doi.org/10.1061/JMENEA.MEENG-7429>
- [J8] **Im, J. B.**, Hong, R. L., Joo, M., Zhang, E., & Kim, J. H.* (2025). Visual Fidelity Effects on Occupants' Performance, Mental States, and Emotions in Mixed Reality Environments. *Architectural Science Review*, Taylor & Francis, 1–21. [A&HCI, Q1, 2.58%, IF 1.8] <https://doi.org/10.1080/00038628.2025.2584180>
- [J7] **Im, J. B.**, Joo, M., Lee, K. T., & Kim, J. H.* (2025). A mixed reality approach to identify cognitive performance and mental states in preferred vs. non-preferred individual and collaborative work environments. *Building and Environment*, Elsevier, 274, 112806. [SCIE, Q1, 4.1%, IF 7.1] <https://doi.org/10.1016/j.buildenv.2025.112806>
- [J6] Park, S. J., **Im, J. B.**, Yoon, H. S., & Kim, J. H.* (2025). Data Augmentation Approaches for Estimating Curtain Wall Construction Duration in High-Rise Buildings. *Buildings*, MDPI, 15(4), 583. [SCIE, Q2, 27.57%, IF 3.1]
<https://doi.org/10.3390/buildings15040583>
- [J5] Kim, H. J., Kim, J. H.* , & **Im, J. B.*** (2023). Forecasting offline retail sales in the COVID-19 pandemic period: A case study of a complex shopping mall in South Korea. *Buildings*, MDPI, 13(3), 627. [SCIE, Q2, 40.4%, IF 3.3] <https://doi.org/10.3390/buildings13030627>
- [J4] Lee, K. T., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2023). Identifying improvements priorities in daycare centre space performance from teachers' perspectives: application of importance–performance analysis and structural equation modelling. *Engineering, Construction and Architectural Management*, Emerald, 32(1), 639–672. [SCIE, Q1, 22.3%, IF 3.6] <https://doi.org/10.1108/ECAM-08-2022-0777>
- [J3] Lee, K. T., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2022). Conceptual framework to support personalized indoor space design decision-making: A systematic literature review. *Buildings*, MDPI, 12(6), 716. [SCIE, Q2, 44.1%, IF 2.6] <https://doi.org/10.3390/buildings12060716>

- [J2] Park, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2022). The need for smart architecture caused by the impact of COVID-19 upon architecture and city: A systematic literature review. *Sustainability*, MDPI, 14(13), 7900. [SCIE, Q2, 45.0%, IF 3.2] <https://doi.org/10.3390/su14137900>
- [J1] Nindartin, A., Moon, H. W., Park, S. J., Lee, K. T., **Im, J. B.***, & Kim, J. H.* (2022). Influencing of the Building Energy Policies upon the Efficiency of Energy Consumption: The Case of Courthouse Buildings in South Korea. *Energies*, MDPI, 15(18), 6679 [SCIE, Q3, 60.9%, IF 3.0] <https://doi.org/10.3390/en15186679>

• **REFERRED CONFERENCE**

- [C11] Hong, R. L., **Im, J. B.**, Shim, W. S., Wang, S. H., Ha, W. C., & Kim, J. H.* (2026). Automatic Semantic Parsing of Structural Drawings and Rebar Schedules Based on a Hybrid Deep Learning Framework, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. <https://doi.org/10.22260/ISARC2026/0197>
- [C10] **Im, J. B.**, Hong, R. L., Choi, C. H., Ha, J. E., & Kim, J. H.* (2026). Optimal Transport-based Affective User Requirements Realignment for Automated Minimal Disruptive Spatial Design Revision in BIM Environment, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. <https://doi.org/10.22260/ISARC2026/0251> (Presenter)
- [C9] **Im, J. B.**, Ye, S. J., Zhang, E., Lee, K. M., Lee, K. T., & Kim, J. H.* (2026). Assessing Material and Labor Cost Indices Impacts on Construction Firms' Financial Performance: A Machine Learning Comparison with Feature Engineering, *the 43rd International Symposium of Automation and Robotics in Construction (ISARC 2026)*, Singapore, 22–26 June, 2026. <https://doi.org/10.22260/ISARC2026/0107>
- [C8] Zhang, E., **Im, J. B.**, Xu, L., Park, S. Y., & Kim, J. H.* (2026). The Influence of Window Characteristics on Occupant Well-Being and Perception in Indoor Environments: A Review, *the 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC 2026)*, Los Angeles, CA, 18–22 May, 2026. <https://doi.org/10.1051/e3sconf/202671605006>
- [C7] Zhang, E., Choi, S. W., **Im, J. B.**, Baek, Y. G., Kim, J. H., & Lee, K. T.* (2025). Construction Conflict Resolution Strategies Focused on Two-level Game Theory: A Case Study Analysis, *the 11th International Conference on Construction Engineering and Project Management (ICCEPM 2025)*, pages 152–119, Jeju, South Korea, 17–20 November, 2025. <https://doi.org/10.6106/ICCEPM.2025.0152>
- [C6] Ochirsuren, N., Kim, G. T., **Im, J. B.**, Hong, R. L., Xu, L., & Kim, J. H.* (2025). A BPMN-based Framework for Real-time Rebar Arrangement Inspection on Construction Sites, *the 11th International Conference on Construction Engineering and Project Management (ICCEPM 2025)*, Pages 136–143, Jeju, South Korea, 17–20 November, 2025. <https://doi.org/10.6106/ICCEPM.2025.0136>
- [C5] Hong, R. L., **Im, J. B.**, Zhang, E., Park, S. J., Ochirsuren, N., Beak, Y. G., Kang, S. H., Wang, S., Lee, C. S., Cheong, S. W., & Kim, J. H.* (2025). Development of a Petrochemical Plant Pipe Counting Method Based on Improved Mask R-CNN and Transformer. *the 42nd International Symposium of Automation and Robotics in Construction (ISARC 2025)*, pages 972–979, Montreal, Canada, 28–31 July, 2025. <https://doi.org/10.22260/ISARC2025/0126>
- [C4] **Im, J. B.**, Hong, R.L., Jang, J. H., Zhang, E., Shin, S. H., Joo, M., Lee, K. W., Lee, K. M., Lee, K. H., Park, S. Y., & Kim, J. H.* (2025). Effects of Visual Fidelity on Task Performance and Mental Workload in IVEs. *the 42nd International Symposium of Automation and Robotics in Construction (ISARC 2025)*, pages 909–916, Montreal, Canada, 28–31 July, 2025. <https://doi.org/10.22260/ISARC2025/0118> (Presenter)
- [C3] Jang, J. H., **Im, J. B.***, Zhang, E. L., Joo, M. B., Kang, S. H., & Kim, J. H.* (2024). A Framework for Assessing the Learning Performance and Creativity in Spatial Features by Immersive Virtual Environments. *the 10th International Conference on Construction Engineering and Project Management (ICCEPM 2024)*, pages 621–628, Sapporo, Japan, 29 July – 1 August, 2024. <https://doi.org/10.6106/ICCEPM.2024.0621> (Presenter)
- [C2] Park, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2022). Smart Architecture in Response to Pandemic: A Systematic Literature Review. *the 13th International Symposium on Architectural Interchanges in Asia (ISAIA 2022)*, Wuhan, China, December 1–2, 2022.
- [C1] Lee, K. T., Park, S. J., **Im, J. B.**, & Kim, J. H.* (2022). Hierarchical Organization Structure Framework for Negotiation Consensus in Multinational Joint-Client Projects. *the 13th International Symposium on Architectural Interchanges in Asia (ISAIA 2022)*, pages 133–140, Wuhan, China, December 1–2, 2022.

DOMESTIC PUBLICATION

• REFERRED JOURNAL

- [6] Shin, S. H., Lee, K. M., **Im, J. B.**, & Kim, J. H.* (2026). Factor Analysis on Cost Overrun by Introducing FMEA and Proposal for Improving Approximate Estimation Reliability in Apartment Projects, *Journal of the Architectural Institute of Korea*, 42(3), 339–347. [Scopus] <https://doi.org/10.5659/JAIK.2026.42.3.339>
- [5] Zhang E. L., Jo, S. H., **Im, J. B.**, & Kim, J. H.* (2024). Importance and Performance Analysis on Stakeholders Perceptions against Public Rental Housing Status and Directions to Improve Occupants Satisfaction. *Journal of the Architectural Institute of Korea*, 41(1), 301–312. [Scopus] <https://doi.org/10.5659/JAIK.2025.41.1.301>
- [4] Lee, K. Y., Ding, Z. S., **Im, J. B.**, Park, S. J., & Kim, J. H.* (2024). Investigation of Construction Duration Influence Factors and Comparison of Artificial Intelligence Methods for Estimating Appropriate Construction Duration in Public Sewerage Construction. *Korea Institute of Construction Engineering and Management*, 25(6), 66–74. [KCI] <https://dx.doi.org/10.6106/KJCEM.2024.25.6.066>
- [3] Kim, D. H., Ye, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2023). Developing a Comprehensive Framework for Detailed Facility Planning in Office Building Grade Assessment. *Journal of the Architectural Institute of Korea*, 39(9), 69–80. [Scopus] <https://doi.org/10.5659/JAIK.2023.39.9.69>
- [2] **Im, J. B.**, Hwang, S. J., Jang, J. H., Kim, J. H., & Lee, K. T.* (2022). Introduction of mixed reality on special tree landscape design review phase. *Journal of the Architectural Institute of Korea*, 38(8), 71–81. [Scopus] <https://doi.org/10.5659/JAIK.2022.38.8.71>
- [1] Kim, H. S., **Im, J. B.**, Lee, K. T., Kim, H. J., & Kim, J. H.* (2021). Proposals for Feasibility Study and Optimal Combination of Renewable Energy Generation Systems in Baengnyeongdo and Oedo using HOMER. *Journal of the Architectural Institute of Korea*, 37(12), 347–356. [Scopus] <https://doi.org/10.5659/JAIK.2021.37.12.347>

• REFERRED CONFERENCES

- [5] Kim, D. H., Ye, S. J., **Im, J. B.**, & Kim, J. H.* (2023). Developing a Comprehensive framework for Detailed Facility Planning in Office Building Grade Assessment. *Spring Annual Conference of AIK*, pages 769–772.
- [4] **Im, J. B.**, Ding, Z. S., & Kim, J. H.* (2023). A Case Study of Complex Shopping Mall on Factors Relevant to COVID-19 Affecting Offline Retail Sales, *Spring Annual Conference of AIK*, pages 735–736.
- [3] Lee, K. T., **Im, J. B.**, Ding, Z. S., Hong, R. L., & Kim, J. H.* (2022). An Analysis of User's Emotions and Task Performance in Indoor Space Implemented by Mixed-Reality, *Autumn Annual Conference of AIK*, pages 1051–1054.
- [2] Park, S. J., Lee, K. T., **Im, J. B.**, & Kim, J. H.* (2022). Design of Experiments for Voxel Down-sampling Indoor Point Cloud Data, *Autumn Annual Conference of AIK*, pages 1082–1085.
- [1] **Im, J. B.**, Lee, K. T., Kim, H. J., Nindartin, A., Nour, N., & Kim, J. H.* (2022). Forecasting of Offline Complex Shopping Mall Retail Sales after COVID-19 pandemic, *Autumn Annual Conference of AIK*, pages 1060–1061.

IN-PROGRESS RESEARCH

[In Review] Strategic Pathways for e-Methanol Deployment in the Low-Carbon Transition: Policy Modeling of Cost Competitiveness and Flexibility Requirements in Korea. Submitted to *Energy Strategy Reviews*.

[In Review] Beyond Viewing: How Installation Artwork Design Modality Shapes Visitor Satisfaction Through Affective and Behavioral Pathways. Submitted to *the Design Journal*.

[In Review] Weather-adaptive automated pipe counting for open-air construction storage: Leveraging synthetic data augmentation and YOLO-Network Family. Submitted to *Scientific Reports*.

[Submitted] Quantifying affect as a design requirement: An affective engineering Approach to modelling multi-dimensional emotional response of residential interiors. Submitted to *Architectural Science Review*.

[Submitted] Optimizing Immersive Virtual Environments for Architectural Design Review: The Impact of Visualization Quality on Occupant-Centric Performance Assessment. Submitted to *Building and Environment*.

[Manuscript Completed] External bargaining and internal ratification in Seoul urban redevelopment through a two-level game theory

[Manuscript Completed] Automated BIM design revision from residents' natural-language emotional feedback

REFERENCE

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